Supplementary Material: How Far Can Simulation Take Vehicle-Based IRI Estimation?

S1. Protocol, deviations and fixes

The evaluation protocol (`evaluation/FINAL_EVALUATION_PROTOCOL.md`) and the model configurations (`config/models.json`) were frozen before any test evaluation. The following decisions were taken afterwards; none was taken in response to a test result unless stated. All are logged in `CHANGELOG.md`.

- *CPH6-HH labels.* LiRA-CD's CPH6-HH profile file and IRI file cannot be aligned: the best Pearson correlation between the 10 m IRI recomputed from the profile and the reported 10 m IRI over shifts of ± 3 km, with and without reversal, is 0.13, while two control files align at zero shift with $r = 1.000$ (`evaluation/cph6hh_alignment.py`). The pair fails the a-priori consistency rule (10 m agreement $r \geq 0.95$); all labels are therefore recomputed from the profiles, which carry GPS.
- *Skipped passes.* 11 LiRA-CD passes lack accelerometer or CAN speed data (9 contain GPS only, 2 have no speed signal) and were skipped; 2 further passes yield no valid unit. On the CPH1 ring road the chainage wraps around; GPS tracks are split into runs of plausible progression (a data-engineering fix found from unit counts, before any model was evaluated).
- *Simulator realisation.* The simulator's first implementation used a polynomial (second-order-section) realisation that deviated by up to 19 % from the state-space solution; it was replaced by the modal first-order-hold realisation before any simulated data were used. A Runge–Kutta check of the final solver was added to the unit tests later; its first version started the continuous and the discrete system from different initial states, which was a fault of the test only (the simulator removes the initial profile offset).
- *Sensor statistics.* The telematics sensor configuration of the simulator (55–62 Hz, 0.01 g, tilt s.d. 6°) was set from a preliminary inspection of the unlabelled LiRA-CD car files that read the accelerometer rates as 56–260 Hz and the tilt as up to 11°. A complete re-measurement for the paper (`evaluation/lira_sensor_stats.py`) gives nominal rates of 52.6 Hz (66 % of passes) or 333 Hz, 0.01 g quantisation in 81 % of passes and tilts with median 10° (maximum 41°). The simulator was not changed after this finding, because the zero-shot results had already been computed. Instead, a post-hoc sensitivity analysis re-ran the zero-shot experiment on measured profiles with the same training vehicles and two sensor variants (`sim_generate.pymeasuredsensormeasuredmix`, `exp_lira.pyzeroshot_sensor`): with the telematics half of the mixture matched to the statistics above, LightGBM reaches MAE 0.46 [0.43, 0.49] m/km against 0.44 [0.41, 0.47] for the primary configuration in the same run (paired difference +0.02 m/km; vibration index 0.54 vs. 0.57); with a single matched telematics sensor and no smartphone-like half it reaches 0.69 [0.66, 0.71] m/km (urban 0.75, motorway 0.60; primary: 0.63 and 0.20). These results are reported but not used for any primary claim.
- *Robustness experiment.* The first version drew new random passes for every perturbation level; it was re-run with common random numbers (variance reduction). The first version is kept in `results/sim/robustness_v1_independent_draws.csv`.
- *Secondary speed experiment.* Added after observing the speed sweep of estimators trained on synthetic profiles: the same sweep for estimators trained on simulated drives over measured profiles (leave-one-route-out). Reported regardless of outcome.
- *Compute allocation.* Zero-shot CNNs (three seeds, two simulation sets) and few-shot CNNs ($k \in \{1, 5, 20\}$ km and all, three seeds) were limited before their results were known. The "all" few-shot budget has a single draw.
- *Seeds.* Neural-net metrics on LiRA-CD and BeamNG are means of per-seed metrics (inside every bootstrap resample); in the controlled simulation experiments, metrics of the seed-averaged prediction.
- *Figures.* The IRI-validation figure now reports the number of off-scale points (ARAN9000 values above the axis limit); no statistic changed.

S2. Simulator parameters

S3. A property of the earlier study's loss

The earlier study's asymmetric Huber loss reduces to a scalar before Keras applies sample weights. In TensorFlow 2.10, moving a large weight between samples of a batch leaves the gradient unchanged; the weights only rescale the batch loss by their mean. The "two-dimensional inverse-frequency sampling" of the full model therefore did not re-weight individual samples. Our re-implementation reproduces this behaviour.

TABLE S1
PARAMETER RANGES OF THE PARAMETRIC SIMULATOR (SENSITIVITY RANGES; SOURCES IN CONFIG/SIM_RANGES.JSON).

group	parameter	range	unit	source
vehicle	m_s	1000 - 1800	kg	G, Z
vehicle	r_y	1 - 1.3	m	G (pitch radius of gyration ~ sqrt(a*b))
vehicle	L	2.4 - 3	m	G, Z
vehicle	a_frac	0.4 - 0.5	-	front-heavy passenger cars
vehicle	f_s	1 - 1.6	Hz	G, S
vehicle	rear_freq_ratio	1 - 1.2	-	G (flat-ride tuning)
vehicle	zeta	0.2 - 0.45	-	G, S
vehicle	m_u_per_wheel	30 - 50	kg	G
vehicle	k_t_per_tire	150000 - 300000	N/m	G (wheel hop 10-15 Hz)
vehicle	x_s	-0.5 - 1.2	m	sensor between CG and front axle (LiRA readme: near front axle)
sensor: autopi	rate / LSB / noise / tilt SD	55-62 Hz / 0.01 g / 150-400 ug/rtHz / 6.0 deg	-	M, D
sensor: phone	rate / LSB / noise / tilt SD	90-210 Hz / 0.0006 g / 100-300 ug/rtHz / 12.0 deg	-	D (smartphone IMUs); cradle resonance range is an assumption, treated as sensitivity
speed	base speed	20-110	km/h	design choice (independent of roughness)

S4. ADDITIONAL RESULTS

Table S2 gives the real-car results per held-out route, Table S3 all estimators trained only on simulated data, and Table S4 all controlled tests. Tables S5 and S6 give pass-level results and the 1 km-block bootstrap. Table S7 lists all BeamNG results and Table S8 all PVS AUCs. Fig. S1 shows the sensor-robustness experiment and Fig. S2 the effect of the earlier split on R^2.

TABLE S2
SEGMENT-LEVEL MAE (M/KM) AND SPEARMAN ρ PER HELD-OUT LIRA-CD ROUTE.

Model	CPH1 MAE	CPH1 rho	CPH6 MAE	CPH6 rho	M3 MAE	M3 rho	M13 MAE	M13 rho
M0 constant (training median)	2.12	–	2.03	–	1.31	0.06	0.79	–
M1 speed-only (LightGBM on speed features)	1.1	0.48	1.11	0.42	0.31	0.2	0.25	0.45
M2c RMS/v^alpha (alpha fitted on training data)	1.03	0.74	0.86	0.74	0.35	0.86	0.25	0.7
M3 LightGBM (all features)	0.6	0.88	0.7	0.84	0.13	0.9	0.15	0.72
M4 MLP	0.6	0.88	0.71	0.83	0.14	0.89	0.16	0.72
M5 1D-CNN	0.75	0.87	0.7	0.78	0.16	0.88	0.17	0.71
M6 inner-validation-selected model	0.6	0.88	0.71	0.83	0.14	0.89	0.16	0.72

TABLE S3
ESTIMATORS TRAINED ONLY ON SIMULATED DATA, EVALUATED ON ALL LIRA-CD ROUTES (SEGMENT LEVEL, CLUSTER-BOOTSTRAP 95% CIS).

Model	MAE (m/km)	Spearman	kappa	units	clusters
sim[iso] physics index (alpha fitted in sim)	0.69 [0.67, 0.72]	0.91 [0.89, 0.91]	0.68	1678	1678
sim[iso] LightGBM	0.71 [0.69, 0.74]	0.92 [0.91, 0.93]	0.7	1678	1678
sim[iso] 1D-CNN	0.58 [0.56, 0.61]	0.94 [0.93, 0.94]	0.77	1678	1678
sim[p79] physics index (alpha fitted in sim)	0.57 [0.54, 0.61]	0.94 [0.93, 0.95]	0.87	1678	1678
sim[p79] LightGBM	0.44 [0.41, 0.47]	0.95 [0.95, 0.96]	0.91	1678	1678
sim[p79+iso] physics index (alpha fitted in sim)	0.57 [0.55, 0.60]	0.94 [0.93, 0.95]	0.84	1678	1678
sim[p79+iso] LightGBM	0.46 [0.43, 0.49]	0.95 [0.95, 0.96]	0.9	1678	1678
sim[p79+iso] 1D-CNN	0.49 [0.46, 0.52]	0.93 [0.92, 0.94]	0.88	1678	1678
sim[p79-fixedvehicle] physics index (alpha fitted in sim)	0.74 [0.70, 0.78]	0.94 [0.94, 0.95]	0.81	1678	1678
sim[p79-fixedvehicle] LightGBM	0.63 [0.60, 0.65]	0.95 [0.95, 0.96]	0.81	1678	1678
sim[p79-idealsensor] physics index (alpha fitted in sim)	0.63 [0.60, 0.66]	0.94 [0.94, 0.95]	0.82	1678	1678
sim[p79-idealsensor] LightGBM	0.67 [0.64, 0.69]	0.95 [0.94, 0.96]	0.74	1678	1678
BeamNG[iri_raw] LightGBM	1.97 [1.90, 2.03]	-0.21 [-0.25, -0.16]	-0.15	1678	1678
BeamNG[iri_raw] physics index	1.96 [1.92, 1.99]	0.88 [0.87, 0.90]	0.03	1678	1678
BeamNG[iri_corr] LightGBM	1.36 [1.30, 1.43]	0.24 [0.19, 0.29]	0.16	1678	1678
BeamNG[iri_corr] physics index	0.85 [0.80, 0.90]	0.93 [0.93, 0.94]	0.68	1678	1678

S5. REPRODUCTION

All numbers in the paper are LaTeX macros generated from the result files by `make_numbers.py`; figures and tables are generated by `make_figures.py` and `make_tables.py`. The full pipeline is `run_all_experiments.py`; see `README.md` for environments, data download and runtimes.

TABLE S4
CONTROLLED GENERALISATION TESTS IN THE PARAMETRIC SIMULATOR WITH PRE-REGISTERED CRITERIA (UNSEEN VEHICLES: RELATIVE MAE INCREASE $\leq$10% INCL. CI; SPEED: |CHANGE| $\leq$10% INCL. CI).

test	model	result	criterion_met
unseen vehicles (random split)	M2a RMS only (alpha=0)	MAE seen 1.365 -> unseen 1.435; diff +0.070 [+0.022, +0.120]	True
unseen vehicles (random split)	M2b RMS/v^0.5	MAE seen 1.066 -> unseen 1.201; diff +0.135 [+0.097, +0.177]	False
unseen vehicles (random split)	M2c RMS/v^alpha (alpha fitted)	MAE seen 1.042 -> unseen 1.200; diff +0.158 [+0.117, +0.204]	False
unseen vehicles (random split)	M2d legacy exponent 2	MAE seen 2.139 -> unseen 2.097; diff -0.042 [-0.110, +0.025]	True
unseen vehicles (random split)	M3 LightGBM	MAE seen 0.742 -> unseen 0.884; diff +0.142 [+0.113, +0.176]	False
unseen vehicles (random split)	M3 LightGBM (no speed)	MAE seen 1.025 -> unseen 1.036; diff +0.011 [-0.026, +0.049]	True
unseen vehicles (random split)	M5 1D-CNN	MAE seen 0.809 -> unseen 1.004; diff +0.195 [+0.167, +0.224]	False
unseen vehicles (random split)	M5 1D-CNN (no speed)	MAE seen 1.137 -> unseen 1.358; diff +0.222 [+0.188, +0.256]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2a RMS only (alpha=0)	MAE low-f_s 1.669 vs high-f_s 1.423; diff -0.246 [-0.332, -0.161]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2b RMS/v^0.5	MAE low-f_s 1.431 vs high-f_s 1.018; diff -0.413 [-0.513, -0.314]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2c RMS/v^alpha (alpha fitted)	MAE low-f_s 1.407 vs high-f_s 0.971; diff -0.437 [-0.531, -0.345]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M2d legacy exponent 2	MAE low-f_s 2.270 vs high-f_s 2.141; diff -0.128 [-0.219, -0.039]	True
vehicle extrapolation (f_s > 1.25 Hz unseen)	M3 LightGBM	MAE low-f_s 0.819 vs high-f_s 1.285; diff +0.466 [+0.416, +0.515]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M3 LightGBM (no speed)	MAE low-f_s 1.082 vs high-f_s 1.487; diff +0.405 [+0.349, +0.464]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M5 1D-CNN	MAE low-f_s 0.782 vs high-f_s 1.261; diff +0.478 [+0.377, +0.589]	False
vehicle extrapolation (f_s > 1.25 Hz unseen)	M5 1D-CNN (no speed)	MAE low-f_s 1.058 vs high-f_s 1.082; diff +0.024 [-0.038, +0.080]	True
BeamNG: vehicle held out (S2 vs S1)	M0 constant (training median)	MAE seen 7.60 -> unseen 7.60; diff +0.00 [-0.00, +0.01]	True
BeamNG: vehicle held out (S2 vs S1)	M1 speed-only (LightGBM on speed features)	MAE seen 7.49 -> unseen 7.61; diff +0.12 [+0.00, +0.23]	True
BeamNG: vehicle held out (S2 vs S1)	M2a RMS only (alpha=0)	MAE seen 6.68 -> unseen 7.27; diff +0.59 [+0.43, +0.73]	False
BeamNG: vehicle held out (S2 vs S1)	M2b RMS/v^0.5	MAE seen 6.36 -> unseen 6.88; diff +0.52 [+0.37, +0.66]	False
BeamNG: vehicle held out (S2 vs S1)	M2c RMS/v^alpha (alpha fitted on training data)	MAE seen 6.34 -> unseen 6.88; diff +0.54 [+0.40, +0.69]	False
BeamNG: vehicle held out (S2 vs S1)	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	MAE seen 6.50 -> unseen 6.70; diff +0.20 [+0.11, +0.29]	True
BeamNG: vehicle held out (S2 vs S1)	M3 LightGBM (all features)	MAE seen 6.02 -> unseen 7.25; diff +1.24 [+0.96, +1.51]	False
BeamNG: vehicle held out (S2 vs S1)	M3 LightGBM (no speed features)	MAE seen 6.01 -> unseen 7.39; diff +1.37 [+1.10, +1.64]	False
BeamNG: vehicle held out (S2 vs S1)	M5 1D-CNN	MAE seen 8.63 -> unseen 10.95; diff +2.32 [+1.78, +2.88]	False
speed 40->100 km/h (sweep_iso)	M2a RMS only (alpha=0)	rel. change +63.3% [+60.8, +65.5]	False
speed 40->100 km/h (sweep_iso)	M2b RMS/v^0.5	rel. change +13.7% [+11.8, +15.4]	False
speed 40->100 km/h (sweep_iso)	M2c RMS/v^alpha (alpha fitted)	rel. change +3.1% [+1.4, +4.7]	True
speed 40->100 km/h (sweep_iso)	M2d legacy exponent 2	rel. change -52.0% [-52.5, -51.5]	False
speed 40->100 km/h (sweep_iso)	M3 LightGBM	rel. change +0.6% [-1.1, +2.5]	True
speed 40->100 km/h (sweep_iso)	M3 LightGBM (no speed)	rel. change +35.7% [+31.9, +38.4]	False
speed 40->100 km/h (sweep_iso)	M5 1D-CNN	rel. change -4.4% [-5.9, -2.5]	True
speed 40->100 km/h (sweep_iso)	M5 1D-CNN (no speed)	rel. change +22.1% [+19.9, +24.2]	False
speed 40->100 km/h (sweep_iso)	raw RMS of a_z (no model, not an IRI)	rel. change +82.5% [+79.1, +85.4]	False
speed 40->100 km/h (sweep_p79)	M2a RMS only (alpha=0)	rel. change +166.7% [+158.7, +171.4]	False
speed 40->100 km/h (sweep_p79)	M2b RMS/v^0.5	rel. change +96.1% [+89.7, +99.9]	False
speed 40->100 km/h (sweep_p79)	M2c RMS/v^alpha (alpha fitted)	rel. change +78.3% [+72.5, +81.8]	False
speed 40->100 km/h (sweep_p79)	M2d legacy exponent 2	rel. change -32.0% [-33.5, -31.1]	False
speed 40->100 km/h (sweep_p79)	M3 LightGBM	rel. change +60.2% [+50.4, +67.7]	False
speed 40->100 km/h (sweep_p79)	M3 LightGBM (no speed)	rel. change +92.2% [+62.2, +117.4]	False
speed 40->100 km/h (sweep_p79)	M5 1D-CNN	rel. change +25.2% [+22.5, +27.4]	False
speed 40->100 km/h (sweep_p79)	M5 1D-CNN (no speed)	rel. change +73.1% [+47.9, +94.2]	False
speed 40->100 km/h (sweep_p79)	raw RMS of a_z (no model, not an IRI)	rel. change +235.0% [+223.2, +242.3]	False
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M2c RMS/v^alpha (alpha fitted)	rel. change +7.4% [+5.2, +10.0]	False
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M3 LightGBM	rel. change -0.9% [-4.0, +2.0]	True
speed 40->100 km/h (sweep_p79_trained_on_p79 (LORO))	M3 LightGBM (no speed)	rel. change +35.6% [+16.7, +51.4]	False

TABLE S5
REAL CARS (LIRA-CD), LEAVE-ONE-ROUTE-OUT, PASS LEVEL (EVERY SEGMENT-PASS IS A UNIT; CIS FROM A CLUSTER BOOTSTRAP OVER SEGMENTS).

Model	MAE (m/km)	Spearman	kappa	units	clusters
M0 constant (training median)	1.59 [1.53, 1.64]	-0.69 [-0.71, -0.66]	-0.63	16293	1678
M1 speed-only (LightGBM on speed features)	0.72 [0.69, 0.75]	0.73 [0.71, 0.74]	0.73	16293	1678
M2a RMS only (alpha=0)	1.44 [1.39, 1.49]	-0.40 [-0.43, -0.37]	-0.32	16293	1678
M2b RMS/v^0.5	1.14 [1.09, 1.18]	0.14 [0.11, 0.18]	0.17	16293	1678
M2c RMS/v^alpha (alpha fitted on training data)	0.72 [0.69, 0.76]	0.87 [0.86, 0.88]	0.72	16293	1678
M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	0.74 [0.71, 0.78]	0.87 [0.86, 0.87]	0.71	16293	1678
M3 LightGBM (all features)	0.44 [0.42, 0.46]	0.91 [0.90, 0.91]	0.86	16293	1678
M3 LightGBM (no speed features)	0.72 [0.69, 0.75]	0.73 [0.72, 0.75]	0.66	16293	1678
M4 MLP	0.43 [0.41, 0.45]	0.92 [0.91, 0.92]	0.87	16293	1678
M5 1D-CNN	0.48 [0.46, 0.51]	0.91 [0.90, 0.91]	0.85	16293	1678
M5 1D-CNN (no speed input)	0.59 [0.56, 0.62]	0.84 [0.83, 0.85]	0.78	16293	1678
M6 inner-validation-selected model	0.43 [0.41, 0.45]	0.92 [0.91, 0.92]	0.87	16293	1678

TABLE S6
REAL CARS (LIRA-CD), SEGMENT LEVEL WITH 95% CIS FROM A CLUSTER BOOTSTRAP OVER 1 KM BLOCKS (SENSITIVITY ANALYSIS).

Model	MAE (m/km)	Spearman	kappa	units	clusters
M0 constant (training median)	1.67 [1.53, 1.81]	-0.67 [-0.71, -0.62]	-0.54	1678	172
M1 speed-only (LightGBM on speed features)	0.74 [0.65, 0.84]	0.80 [0.77, 0.82]	0.79	1678	172
M2a RMS only (alpha=0)	1.51 [1.38, 1.63]	-0.36 [-0.42, -0.28]	-0.25	1678	172
M2b RMS/v^0.5	1.13 [1.03, 1.24]	0.29 [0.19, 0.39]	0.29	1678	172
M2c RMS/v^alpha (alpha fitted on training data)	0.67 [0.60, 0.75]	0.91 [0.89, 0.92]	0.8	1678	172
M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	0.73 [0.65, 0.82]	0.91 [0.88, 0.92]	0.77	1678	172
M3 LightGBM (all features)	0.42 [0.36, 0.48]	0.95 [0.94, 0.96]	0.91	1678	172
M3 LightGBM (no speed features)	0.65 [0.58, 0.73]	0.87 [0.85, 0.89]	0.78	1678	172
M4 MLP	0.42 [0.36, 0.48]	0.95 [0.94, 0.96]	0.91	1678	172
M5 1D-CNN	0.48 [0.41, 0.55]	0.94 [0.92, 0.95]	0.89	1678	172
M5 1D-CNN (no speed input)	0.54 [0.47, 0.62]	0.92 [0.90, 0.93]	0.86	1678	172
M6 inner-validation-selected model	0.42 [0.36, 0.48]	0.95 [0.94, 0.96]	0.91	1678	172

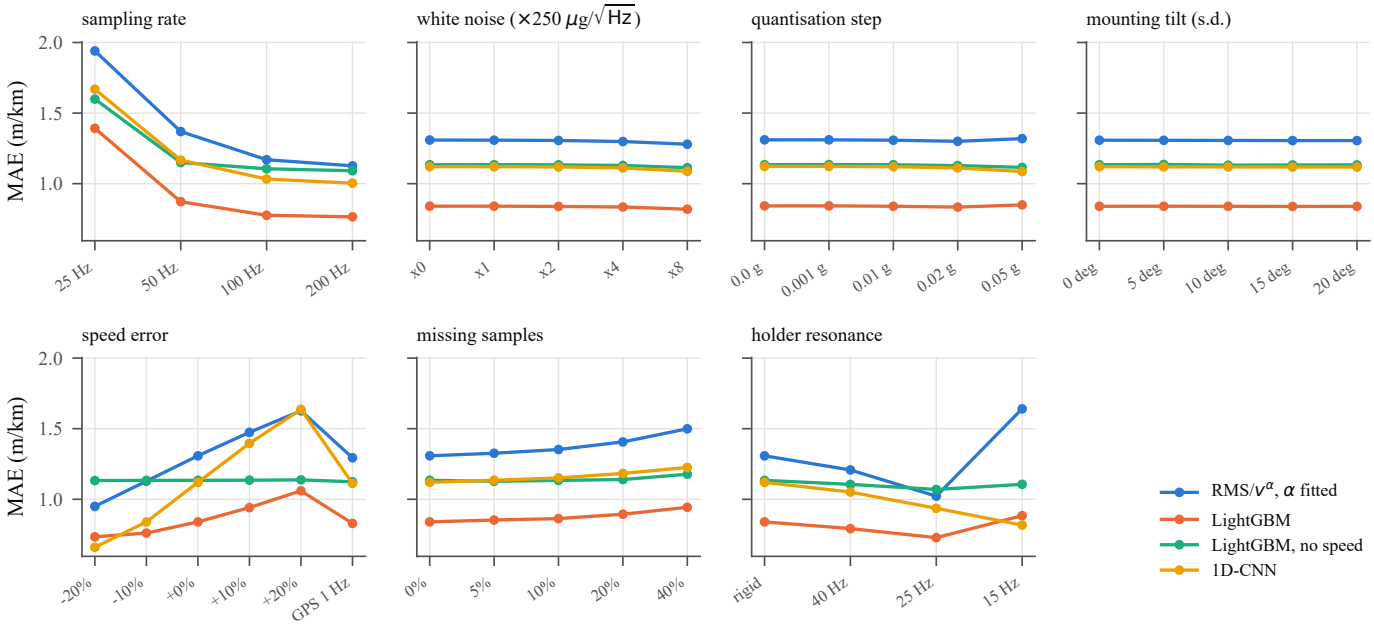


Fig. S1. Sensor robustness in simulation: estimators trained on synthetic profiles, evaluated on 12 held-out synthetic routes driven by 6 unseen vehicles with the telematics sensor configuration; one property changed at a time (common random numbers).

TABLE S7
BEAMNG CORPUS: ALL ESTIMATORS AND SPLITS (SEGMENT LEVEL, CLUSTER-BOOTSTRAP 95% CIS OVER 100 M LOCATION CELLS).

Split	Label	Model	MAE (m/km)	Spearman	R2
S3: earlier split (val. and test on test track)	legacy	M0 constant (training median)	2.68 [2.04, 3.39]	–	-0.07 [-0.19, -0.00]
S3: earlier split (val. and test on test track)	legacy	M1 speed-only (LightGBM on speed features)	2.20 [1.65, 2.85]	0.47 [0.32, 0.59]	0.14 [-0.01, 0.26]
S3: earlier split (val. and test on test track)	legacy	M2a RMS only (alpha=0)	2.25 [1.71, 2.85]	0.67 [0.55, 0.75]	0.11 [-0.25, 0.34]
S3: earlier split (val. and test on test track)	legacy	M2b RMS/v^0.5	2.10 [1.60, 2.65]	0.71 [0.60, 0.79]	0.23 [-0.04, 0.42]
S3: earlier split (val. and test on test track)	legacy	M2c RMS/v^alpha (alpha fitted on training data)	2.03 [1.56, 2.55]	0.72 [0.62, 0.79]	0.30 [0.16, 0.41]
S3: earlier split (val. and test on test track)	legacy	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	2.02 [1.55, 2.53]	0.68 [0.57, 0.76]	0.29 [0.20, 0.36]
S3: earlier split (val. and test on test track)	legacy	M3 LightGBM (all features)	1.88 [1.45, 2.36]	0.74 [0.63, 0.81]	0.40 [0.18, 0.53]
S3: earlier split (val. and test on test track)	legacy	M3 LightGBM (no speed features)	1.97 [1.50, 2.48]	0.72 [0.60, 0.80]	0.35 [0.20, 0.50]
S3: earlier split (val. and test on test track)	legacy	M4 MLP	1.79 [1.34, 2.27]	0.76 [0.67, 0.83]	0.41 [0.31, 0.54]
S3: earlier split (val. and test on test track)	legacy	M5 1D-CNN	2.39 [1.47, 3.58]	0.75 [0.65, 0.82]	-3.28 [-14.77, 0.43]
S3: earlier split (val. and test on test track)	legacy	M5 1D-CNN (no speed input)	3.20 [1.69, 5.72]	0.69 [0.59, 0.77]	-17.73 [-75.41, 0.35]
S3: earlier split (val. and test on test track)	legacy	Legacy recipe (calibrated on val)	1.67 [1.29, 2.11]	0.80 [0.72, 0.85]	0.44 [0.02, 0.68]
S3: earlier split (val. and test on test track)	legacy	Legacy recipe (uncalibrated)	330.85 [1.43, 1043.50]	0.80 [0.72, 0.85]	-5902105.16 [-24835354.32, 0.63]
S1: leave-one-map-out	legacy	M0 constant (training median)	3.01 [2.61, 3.48]	-0.37 [-0.48, -0.25]	-0.21 [-0.30, -0.16]
S1: leave-one-map-out	legacy	M1 speed-only (LightGBM on speed features)	2.60 [2.26, 3.00]	0.31 [0.22, 0.39]	0.05 [-0.05, 0.13]
S1: leave-one-map-out	legacy	M2a RMS only (alpha=0)	2.33 [1.98, 2.74]	0.56 [0.51, 0.61]	0.10 [0.02, 0.18]
S1: leave-one-map-out	legacy	M2b RMS/v^0.5	2.15 [1.83, 2.52]	0.66 [0.61, 0.70]	0.22 [0.14, 0.31]
S1: leave-one-map-out	legacy	M2c RMS/v^alpha (alpha fitted on training data)	2.10 [1.79, 2.46]	0.67 [0.61, 0.71]	0.26 [0.18, 0.35]
S1: leave-one-map-out	legacy	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	2.13 [1.83, 2.49]	0.64 [0.58, 0.69]	0.28 [0.20, 0.36]
S1: leave-one-map-out	legacy	M3 LightGBM (all features)	2.06 [1.77, 2.41]	0.68 [0.63, 0.72]	0.30 [0.22, 0.38]
S1: leave-one-map-out	legacy	M3 LightGBM (no speed features)	2.07 [1.77, 2.42]	0.67 [0.62, 0.71]	0.30 [0.23, 0.38]
S1: leave-one-map-out	legacy	M4 MLP	1.90 [1.64, 2.22]	0.73 [0.69, 0.77]	0.37 [0.31, 0.43]
S1: leave-one-map-out	legacy	M5 1D-CNN	2.23 [1.79, 2.77]	0.70 [0.66, 0.74]	-3.33 [-13.32, 0.35]
S1: leave-one-map-out	legacy	M5 1D-CNN (no speed input)	2.64 [1.98, 3.60]	0.62 [0.56, 0.67]	-15.92 [-60.71, 0.22]
S1: leave-one-map-out	legacy	Legacy recipe (calibrated on val)	1.99 [1.76, 2.25]	0.66 [0.59, 0.72]	0.45 [0.37, 0.50]
S1: leave-one-map-out	legacy	Legacy recipe (uncalibrated)	251.39 [1.82, 773.08]	0.68 [0.62, 0.73]	-17368673.46 [-65761393.61, 0.37]
S1: leave-one-map-out	raw profile	M0 constant (training median)	7.60 [6.67, 8.67]	-0.25 [-0.36, -0.14]	-0.15 [-0.23, -0.09]
S1: leave-one-map-out	raw profile	M1 speed-only (LightGBM on speed features)	7.49 [6.64, 8.46]	0.21 [0.12, 0.30]	-0.10 [-0.23, 0.00]
S1: leave-one-map-out	raw profile	M2a RMS only (alpha=0)	6.68 [5.90, 7.58]	0.49 [0.42, 0.55]	0.08 [-0.03, 0.17]
S1: leave-one-map-out	raw profile	M2b RMS/v^0.5	6.36 [5.62, 7.23]	0.54 [0.47, 0.60]	0.16 [0.05, 0.25]
S1: leave-one-map-out	raw profile	M2c RMS/v^alpha (alpha fitted on training data)	6.34 [5.60, 7.20]	0.54 [0.46, 0.60]	0.17 [0.06, 0.25]
S1: leave-one-map-out	raw profile	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	6.50 [5.72, 7.40]	0.48 [0.38, 0.56]	0.13 [0.01, 0.22]
S1: leave-one-map-out	raw profile	M3 LightGBM (all features)	6.02 [5.37, 6.73]	0.57 [0.50, 0.63]	0.26 [0.14, 0.35]
S1: leave-one-map-out	raw profile	M3 LightGBM (no speed features)	6.01 [5.38, 6.75]	0.58 [0.51, 0.63]	0.26 [0.15, 0.35]
S1: leave-one-map-out	raw profile	M4 MLP	5.70 [5.15, 6.30]	0.59 [0.52, 0.64]	0.34 [0.19, 0.44]
S1: leave-one-map-out	raw profile	M5 1D-CNN	8.77 [5.82, 14.01]	0.51 [0.44, 0.58]	-110.29 [-387.20, 0.22]
S1: leave-one-map-out	raw profile	M5 1D-CNN (no speed input)	9.75 [5.90, 16.77]	0.52 [0.44, 0.59]	-214.49 [-752.26, 0.13]
S2: leave-one-vehicle-and-map-out	raw profile	M0 constant (training median)	7.60 [6.67, 8.67]	-0.23 [-0.34, -0.12]	-0.15 [-0.23, -0.09]
S2: leave-one-vehicle-and-map-out	raw profile	M1 speed-only (LightGBM on speed features)	7.61 [6.80, 8.55]	0.21 [0.11, 0.29]	-0.11 [-0.25, -0.00]
S2: leave-one-vehicle-and-map-out	raw profile	M2a RMS only (alpha=0)	7.27 [6.54, 8.09]	0.42 [0.36, 0.47]	-0.02 [-0.15, 0.07]
S2: leave-one-vehicle-and-map-out	raw profile	M2b RMS/v^0.5	6.88 [6.18, 7.68]	0.48 [0.41, 0.54]	0.07 [-0.07, 0.17]
S2: leave-one-vehicle-and-map-out	raw profile	M2c RMS/v^alpha (alpha fitted on training data)	6.88 [6.18, 7.69]	0.47 [0.40, 0.52]	0.07 [-0.07, 0.16]
S2: leave-one-vehicle-and-map-out	raw profile	M2d RMS*(v_ref/v)^2 (legacy exponent, gravity removed)	6.70 [5.94, 7.57]	0.45 [0.36, 0.53]	0.09 [-0.04, 0.18]
S2: leave-one-vehicle-and-map-out	raw profile	M3 LightGBM (all features)	7.25 [6.64, 7.90]	0.46 [0.39, 0.51]	0.02 [-0.17, 0.15]
S2: leave-one-vehicle-and-map-out	raw profile	M3 LightGBM (no speed features)	7.39 [6.77, 8.05]	0.45 [0.39, 0.51]	-0.01 [-0.21, 0.12]
S2: leave-one-vehicle-and-map-out	raw profile	M5 1D-CNN	11.06 [8.13, 16.02]	0.43 [0.36, 0.49]	-173.32 [-605.45, -0.78]

TABLE S8
PVS: LEAVE-ONE-ROUTE-OUT LOGISTIC REGRESSIONS, AUC WITH CLUSTER-BOOTSTRAP 95% CIS OVER GPS CELLS.

Task	Predictors	AUC [95% CI]	windows	positives
A: asphalt vs cobblestone+dirt	speed	0.97 [0.94, 0.99]	1070	352
A: asphalt vs cobblestone+dirt	RMS	0.93 [0.91, 0.94]	1070	352
A: asphalt vs cobblestone+dirt	RMS/sqrt(v)	0.99 [0.98, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	model	0.94 [0.90, 0.96]	1070	352
A: asphalt vs cobblestone+dirt	model+speed	0.99 [0.97, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	model+RMS	0.96 [0.95, 0.98]	1070	352
A: asphalt vs cobblestone+dirt	model+speed+RMS	0.99 [0.99, 1.00]	1070	352
A: asphalt vs cobblestone+dirt	speed+RMS	0.99 [0.98, 1.00]	1070	352
B: good vs regular+bad	speed	0.97 [0.95, 0.99]	1070	358
B: good vs regular+bad	RMS	0.93 [0.91, 0.95]	1070	358
B: good vs regular+bad	RMS/sqrt(v)	0.99 [0.98, 1.00]	1070	358
B: good vs regular+bad	model	0.94 [0.90, 0.96]	1070	358
B: good vs regular+bad	model+speed	0.99 [0.97, 1.00]	1070	358
B: good vs regular+bad	model+RMS	0.97 [0.95, 0.98]	1070	358
B: good vs regular+bad	model+speed+RMS	1.00 [0.99, 1.00]	1070	358
B: good vs regular+bad	speed+RMS	0.99 [0.98, 1.00]	1070	358
C: bad vs regular (cobblestone+dirt only)	speed	0.75 [0.67, 0.82]	352	72
C: bad vs regular (cobblestone+dirt only)	RMS	0.23 [0.15, 0.32]	352	72
C: bad vs regular (cobblestone+dirt only)	RMS/sqrt(v)	0.43 [0.35, 0.51]	352	72
C: bad vs regular (cobblestone+dirt only)	model	0.41 [0.33, 0.50]	352	72
C: bad vs regular (cobblestone+dirt only)	model+speed	0.75 [0.67, 0.82]	352	72
C: bad vs regular (cobblestone+dirt only)	model+RMS	0.51 [0.42, 0.58]	352	72
C: bad vs regular (cobblestone+dirt only)	model+speed+RMS	0.80 [0.72, 0.86]	352	72
C: bad vs regular (cobblestone+dirt only)	speed+RMS	0.78 [0.70, 0.84]	352	72

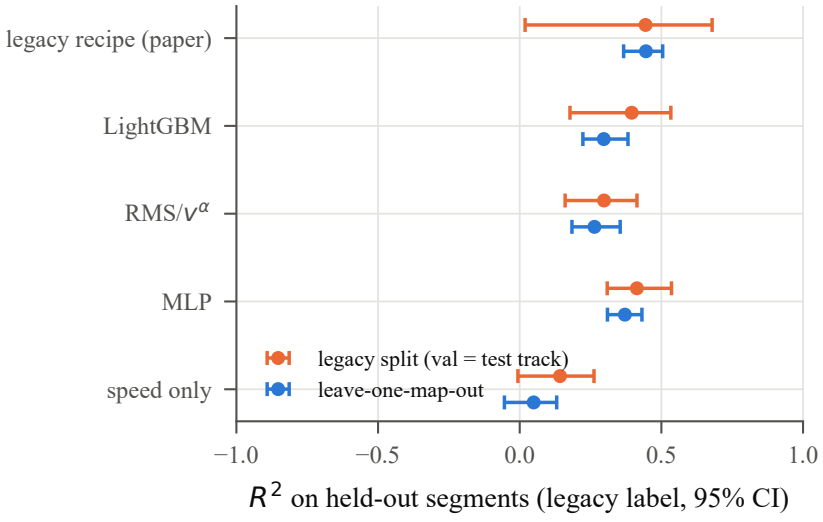


Fig. S2. BeamNG corpus, earlier label definition: R^2 on held-out segments under the earlier split (validation and test trips on the same track) and under leave-one-map-out (95 % cluster-bootstrap CIs).